

Tokenization

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Machine learning is not used to create tokens

It's a simple process

SolidGoldMagikarp became a token

Was removed along with other glitch tokens

Don't want junk data because it makes it harder to learn relationships and increases dimensionality

When we enter something to an LLM, the actual thing that gets entered is a vector of tokens

SolidGoldMagikarp was only related to numbers in the counting Reddit within the training data

This made the embedding for it really bad, there is nothing to draw understanding from

Words are not tokens

Byte Pair Encoding

Add all characters

Pairs most frequent combinations of characters together

We set k, run until you hit k

Attention:

Builds relationships and meaning between words

Takes tokens that are not directly next to each other and combines their meaning

The larger your vocab, the more time/resource expensive it is

Stemming

Just chops off the ends of words

Lemmatization

Uses actual rulesets to determine the root word/meaning of words, i.e. better becomes good

Con is that you need to have a dictionary

Modern models don't use stemming and lemmatization much anymore

AI photo tags accidentally tagged a black man as a gorilla

Google removed the models ability to classify gorillas